Joshua Hatfield

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LinkedIn • GitHub • Website

Career Objective

Seeking to pursue a Ph.D. in Computer Science focused on reinforcement learning, virtual reality, and human-centered AI, with the goal of creating adaptive simulation environments that enhance training, safety, and decision-making in complex systems

Research Interests

- Procedural Content Generation & Generative AI
- Reinforcement Learning & Human-Centered AI
- Immersive Analytics & XR Simulation
- Embedded & Secure Systems
- Digital Twins & IoT Systems
- Data Science

Education

Colorado School of Mines, *Golden, CO*

(GPA: TBD) Ph.D. in Computer Science — *Expected May 2030*
Focus on HCI

Marshall University*, Huntington, WV*

(GPA: **4.0**) M.S. in Computer Science — *Aug 2026*

(GPA: **3.9**) B.S. in Computer Science with Honors — *May 2025*
Minor: Mathematics

Work Experience

Teaching Assistantship – Colorado School of Mines — *Starting Aug 2026*

- *To provide guidance for students at Mines*

IT/Data Analyst – Best Virginia Heating and Cooling LLC — *Jan 2026 – Present*

- *Managed* computer and software systems for 70+ employees.
- *Streamlined* data collection, extraction, and analysis to better inform field techs.
- *Directed* price book for 8-figure company across different departments.

(Graduate) Research Assistant – Marshall University — *Aug 2023 – May 2026*

- *Designed* VR-based training simulations integrating reinforcement-learning feedback loops to enhance user performance and retention.
- *Synthesized* technical documentation detailing system logic, architectural choices, testing conditions, and performance outcomes.
- *Awarded* the Creative Discovery and Research Award (2024 and 2026) for innovation in cross-disciplinary research pertaining to social cost-based migration in Covid-19 simulation and workforce training

Future Computer Summer Intern – National Security Agency (NSA) — *Jun – Aug 2024*

- *Engineered* an FPGA-based True Random Number Generator achieving 99%+ NIST-verified randomness for secure cryptographic applications.
- *Authored* comprehensive validation reports and presented outcomes to agency technical cohort.

- *Contributed* to the design of reproducible embedded-security frameworks and entropy verification systems.

Research Experience for Undergraduates (REU)– University of Louisville *Jun – Aug 2023*

- *Optimized* Tensor Virtual Machine (TVM) inferences across variable CPU frequencies for heterogeneous computing environments.
- *Conducted* benchmark and performance analysis to improve deployment efficiency of ML models on constrained devices.
- *Delivered* findings during the REU Research Symposium, receiving commendation for clarity and rigor.

Teaching & Mentorship

Teaching Assistant – Marshall University *Aug – Dec 2023*

- *Guided* 40+ students in algorithmic design, graded assignments, and supported course delivery.

Publications & Presentations

Hatfield, J., Wahjudi P., Narman H., Chowdhury S., Alzarrad A. (2026) *State-Aware Adaptive Feedback and Performance Telemetry in Virtual Reality Industrial Maintenance Training.* Thesis at Marshall University for M.S. in Computer Science

Hatfield J., Narman H., Chowhury S., Alzarrad A. (2026) *State-Aware Adaptive Feedback in Virtual Reality for Safety-Critical Infrastructure Training.* Publication accepted at Global Humanitarian Technology Conference (GHTC).

Hatfield J., Narman H., Chowhury S., Alzarrad A. (2025) *A Scenario-Driven Virtual Reality Hydroelectric Training Environment with Fine-Grained Telemetry for Performance Evaluation.* Publication accepted at ROSE IEEE

Hatfield J., Narman H., Chowhury S., Alzarrad A. (2025) *Assessing Procedural Content Generation in Reinforcement Learning Prompt-Based Feedback via Adaptive Virtual Reality Industrial Environments for Workforce Training.* Publication accepted at ROSE IEEE

Hatfield J., Chowdhury S., Ammar Alzarrad (2026) *Modeling Community Resilience Under Prolonged Disruption: An Agent-Based Framework Integrating Social Connectivity, Migration, and Policy-Driven Allocation.* Work which was awarded the Creative Discovery and Research Award in 2024 and published to MDPI jAnalytics journal in 2026.
Hatfield J., Serdyuk K., Strots B., Baker N., Simon T., Gauthier D., Lui W. (2024) *An Analysis of Entropy and Randomness as it Relates to True Random Number Generation.* Technical Report presented to NSA Future Computer Summer Internship Cohort.

Hatfield, J., Baidya S., Kutukcu B. (2023). *Optimizing Machine Learning Models Using TVM.* Presented at REU Research Symposium, University of Louisville.

Selected Projects

AI-Driven VR Simulation Framework: Designed a reinforcement-learning feedback system in Unreal Engine 5 to train users in hydroelectric maintenance.

FPGA-Based TRNG: Implemented and validated a hardware entropy generator for cryptographic systems using NIST SP 800-22 standards.

GameShelf Web Platform: Developed a cloud-based game catalog using Azure SQL, Node.js, and React, supporting user reviews and social connectivity.

Facial Recognition Attendance System: Built real-time recognition and logging across multi-camera networks with OpenCV and Python.

Scholarships

Logan County Charitable and Educational Foundation Grant	2025
Dorsey Lee Spaulding Family Scholarship	2021
John Laidley Scholarship	2021
Promise Scholarship	2021
Honors College Scholarship	2021
Federal Pell Grant	2021
West Virginia Higher Education Grant	2021
United States Senate Youth Program Scholarship	2020

Honors & Awards

Creative Discovery and Research Award, Marshall University	2026
Summa Cum Laude	2025
ASEE North Central Student Service Award	2025
National Cyber League: Top 500	2025
ALEF Scholar, Appalachian Leadership and Education Foundation	2021-2025
Creative Discovery and Research Award, Marshall University	2024
U.S. Senate Youth Delegate	2020

Leadership & Service

President, Marshall University Technology Association	2024–2025
Member, Engineering Student Advisory Council	2024-2025

Technical Skills

Programming: Python, C++, C, Java, C#, SQL, Verilog, JavaScript, Go, Lua
Frameworks & Tools: Unreal Engine, TVM, React, Node.js, Azure, Git, Linux, FPGA Design, Service Titan, Contractor Commerce, ServiceLite,

Professional Affiliations

- Institute of Electrical and Electronics Engineers (IEEE), Student Member
- Marshall University Technology Association, President
- Appalachian Education Leadership Foundation, Alumni

Hyperlinks

LinkedIn: https://www.linkedin.com/in/joshua-hatfield-675474192/

GitHub: https://github.com/sirjoshies

References: Available upon request